

Design a ML project

- Choose a topic you are interested to explore
 - Depend on the data you have, the area you want to work on
 - The problem is practical, analytical, and makes sense (use google)
 - The target (the things you want to predict) can be quantified and correlated with your features (use common sense)
 - The performance can be evaluated with certain metrics. The results can be reproduced
 - The workload is manageable – 2 months for this course project. Don't work on a problem which is too hard or a dataset which is too complex

Agenda (1)

- Form a team (**3-5 students**), utilize canvas to find teammates
- Proposal (**due TBD**)
 - Pick a project idea to work on
 - Title, project team members (name and ID), **300-500 words** description of what you plan to do
 - Motivation: What problem are you tackling?
 - Data: What data are you learning from?
 - Method: What techniques are you planning to apply? What challenge are you trying to solve?
 - Experiments: What experiments are you planning to run? How do you plan to evaluate your framework?

Agenda (2)

- Final Report (**due TBD**)
 - At most **10 pages**
 - **1. Abstract (optional) and Introduction**
 - Motivation for pursuing this problem
 - Background of the problem
 - What is the input and output of your framework (explicitly with features and predictions)
 - What is the models you use and why
 - What is the conclusion and insights you have got
 - **2. Related Work**
 - What is the challenge of your study problem?
 - What is the existing method for your study problem (papers, online search results, less than 5)
 - What is their strength and weakness
 - How is your work similar or different from them?

Agenda (3)

- Final Report (**due TBD**)
 - **3. Dataset and Formation**
 - Describe your dataset, how it is collected, feature list, data statistics (e.g., feature/target distribution, male/female, number of subjects)
 - What is the data preprocessing you did (normalization, missing value, feature selection, sampling ...)?
 - What are the inputs and output?
 - What is the data size (training/validation/testing)?
 - **4. Method**
 - Your learning algorithms
 - Describe the model, its mechanism, how it works in your scenario
 - Learning objective in formal language and math equations (refers to textbook and lecture notes)
 - What is the strength and weakness of your model?

Agenda (4)

- Final Report (**due TBD**)
 - **5. Experiments:**
 - What is the setting of your experiments and how you find it (settings for your model like learning rate, iteration, batch size, cross-validation)
 - Describe your training procedure
 - Analyze if you had overfitting problem and how to handle it
 - What are your baselines and why?
 - List and explain your evaluation metrics (accuracy, precision, AUC...)
 - Show results in tables and visualization
 - Show qualitative insights from your results
 - Discuss your results and the future work (optional)
 - **6. Conclusion:** what have you done, what did you find (short)
 - **7. References:** The existing work you had reviewed (less than 5)